

Designing an *In Vitro* Neural Network Model

Disease: Neuropathetic Pain

1.

Affected areas:

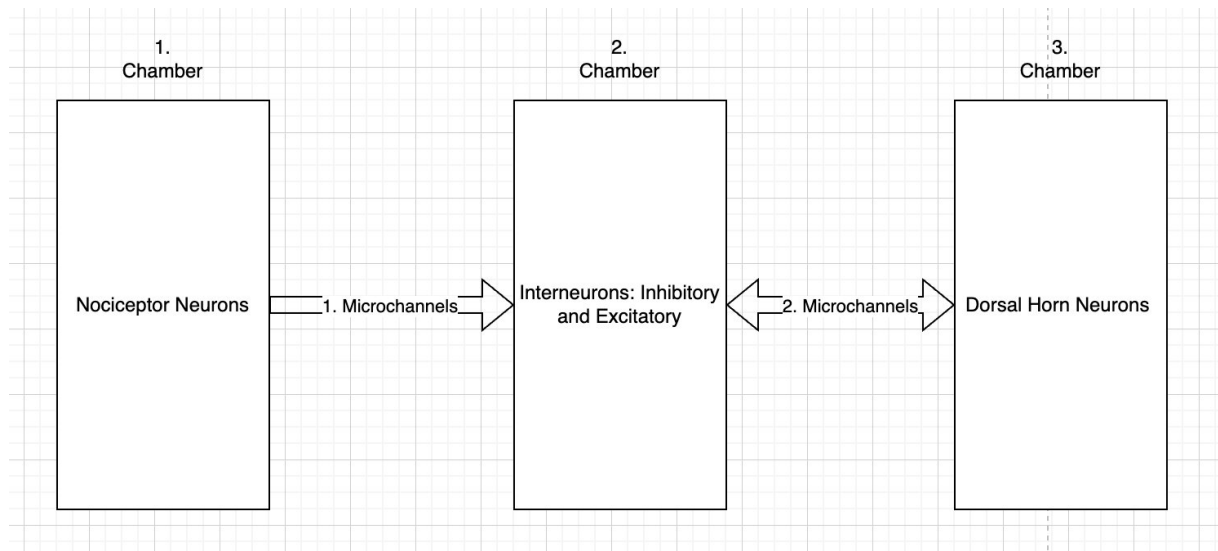
- Peripheral Nociceptor Neurons
- Spinal Cord Dorsal Horn Neurons
- Brain Pain Pathway Neurons

Circuit Connectivity:

- Nociceptors take some input
- Most commonly spinal cord recurrent circuitry amplifies pain signals
- Amplified pain signal is sent as output to the brain

2.

The circuit consists of three chambers. The first chamber contains nociceptor neurons, including their nociceptive terminals, which receive excitatory input, and their cell bodies (soma). The axons of these nociceptor neurons extend through microchannels into the second chamber. The second chamber contains both inhibitory and excitatory interneurons. Their axons project through additional microchannels into the third chamber, which houses dorsal horn neurons. The axons of the dorsal horn neurons can also extend back into the second chamber, enabling the formation of recurrent circuits.



Nociceptor neurons project into dorsal horn circuits where excitatory and inhibitory interneurons regulate signal transmission. The three-chamber design recreates this functional structure by separating nociceptors, interneurons, and dorsal horn neurons while guiding axonal projections through microchannels. This organization preserves the natural flow of nociceptive signaling and allows targeted investigation of inhibitory modulation, a key mechanism involved in neuropathic pain.

3.

The disease state can be modeled in two ways: by directly stimulating the nociceptor neurons or by electrically stimulating their axons. Electrical stimulation provides precise experimental control, allowing different patterns of neuronal activation to be simulated by varying the strength, frequency, and waveform of the applied signal. This approach enables the modeling of abnormal nociceptor activity that is characteristic of neuropathic pain, such as increased excitability or spontaneous firing. By adjusting the stimulation parameters, different pathological excitation conditions can be reproduced and their effects on the downstream interneuron and dorsal horn circuitry can be systematically investigated.

4.

MEAs can be used to monitor interneuron activity within the circuit. MEAs enable simultaneous extracellular recordings from multiple neurons, allowing analysis of firing rates and spike patterns. These measurements provide insight into the excitation levels of the interneurons and how they change in response to nociceptor stimulation. For example, one would study to which effect excitatory interneurons overexcite and how degenerate inhibitory interneurons become.

5.

Modulation through strength, frequency, waveform and chemical additives. Potential treatments can be tested by adding certain chemicals and studying the effect.